THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The radii of two circles are 6 cm and 8 cm. The radius of the circle whose area equals the sum of their areas is: 1

(a) 14 cm **(b)** 12 cm **(c)** 10 cm **(d)** 2 cm

2. The value of $\sin 15^\circ / \cos 75^\circ$ is: 1

(a) 1 **(b)** 2 **(c)** 15 **(d)** 0

3. A storm breaks a tree at a height of 4 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. The original height of the tree is: 1

(a) $12\sqrt{3}$ m **(b)** 24 m **(c)** 12 m **(d)** 8 m

4. A tangent to a circle of radius 9 cm has length 40 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 41 cm **(b)** 31 cm **(c)** 82 cm **(d)** 49 cm

5. The minute hand of a clock is 21 cm long. The area swept by the minute hand in 5 minutes is: 1

(a) 115.5 cm² **(b)** 231 cm² **(c)** 1386 cm² **(d)** 11 cm²

6. $(2\sin\theta)^2 + (2\cos\theta)^2 =$ 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 1

7. From the top of a building 7 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. The height of the tower is: 1

(a) $7 + 7\sqrt{3}$ m **(b)** $14 + 14\sqrt{3}$ m **(c)** $7\sqrt{3}$ m **(d)** 14 m

8. In $\triangle ABC$, right-angled at B, $AB = 6$ cm and $BC = 8$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 4 cm **(b)** 5 cm **(c)** 2 cm **(d)** 3 cm

9. A horse is tied at a corner of a large square field with a rope 7 m long. The area of the field it can graze is: 1

(a) 38.5 m² **(b)** 11 m² **(c)** 154 m² **(d)** 77 m²

10. If $\sec\theta = \frac{5}{3}$ then $\cos\theta =$ 1

(a) \frac{4}{3} **(b)** \frac{3}{5} **(c)** \frac{4}{5} **(d)** \frac{5}{3}

11. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is 10 m above the level of the person's eyes and the eyes are 10 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) 60° **(b)** 15° **(c)** 30° **(d)** 45°

12. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (3x + 5)$ cm and $QB = (2x + 9)$ cm. The length of QA is: 1

(a) 4 cm **(b)** 19 cm **(c)** 17 cm **(d)** 21 cm

13. The area of a sector of a circle of radius 21 cm is 231 cm^2. The central angle of the sector is: 1

(a) 61° **(b)** 120° **(c)** 60° **(d)** 90°

14. If $\tan A = \frac{12}{5}$, then $\sin A =$ 1

(a) \frac{5}{13} **(b)** \frac{12}{5} **(c)** \frac{13}{12} **(d)** \frac{12}{13}

15. From a point on level ground, 21 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) $21\sqrt{3}$ m **(b)** 21 m **(c)** $7\sqrt{3}$ m **(d)** 7 m

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A ladder 14 m long leans against a vertical wall and makes an angle of 60° with the level ground. Find the distance of the foot of the ladder from the wall. 2

17. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 40^\circ$. Find the $\angle OAB$. 2

18. A wire in the form of a circle of radius 14 cm is bent into a square. Find the area of the square. 2

19. Evaluate: $\dfrac{\tan 21^\circ}{\cot 69^\circ}$. 2

20. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is $10\sqrt{3}$ m above the level of the person's eyes and the eyes are 10 m horizontally from the pole. Find the angle of elevation of the top of the pole. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A sector of a circle of radius 14 cm has central angle 90°. Find the perimeter of the sector. 3

22. Prove that: $(1 - \cos^2\theta)\csc^2\theta = 1$. 3

23. From the top of a tower 75 m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 45°. Find the distance between the two objects. 3

24. Two concentric circles have radii 5 cm and 3 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 3

OR

An umbrella has 8 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 28 cm. Find the area between two consecutive ribs. 3

25. If $\tan\theta = \dfrac{15}{8}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 34 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. If $\tan A = \frac{4}{3}$, evaluate $\frac{4\sin A + 3\cos A}{4\sin A + 3\cos A}$. 5

OR

A statue 2 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. Find the height of the pedestal. 5

27. In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 6$ cm and $BC = 8$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). 5

OR

A brooch is a circular piece of silver wire of diameter 35 mm. The wire is also used for 5 equal diameters that divide the brooch into 10 sectors. Find the total length of wire required (in mm). 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND